

CN-PRACTICAL 1: CONFIGURATION LAN SETUP

Aim: To design, configure, and verify a Local Area Network (LAN) using a **Star Topology** and perform network diagnostics using Command Line Interface (CLI) tools.

Configuring LAN setup

- a. Planning and Setting IP Networks
- b. Configuring subnet
- c. Using Linux Terminal or Windows CMD, execute the following networking commands and note the output: ping, traceroute, netstat, arp, ipconfig, Getmac, hostname, NSLookUp, pathping, SystemInfo

1. Star Topology

The Star Topology is the physical arrangement used for our Local Area Network (LAN) setup. It is the most common topology in modern networks due to its reliability and ease of management.

Definition

The Star Topology connects every network device (node) directly to a single, central connecting device, known as a Switch or Hub. This connection creates a network that geometrically resembles a star.

Characteristics

- **Centralized Control:** All data must pass through the central Switch.
- **Dedicated Links:** Each device has its own dedicated cable segment connecting it to the switch.
- **High Fault Tolerance:** If a cable connecting a single PC fails, only that PC is disconnected. The rest of the network continues to function normally.
- **Single Point of Failure:** If the central Switch fails, the entire network is shut down.
- **Scalability:** It is easy to add or remove devices without interrupting the other connections.

2. Procedure: LAN Configuration and Command Execution

A. Network Setup and IP Planning

1. **Place Devices:** Drag one 2960-24TT Switch (or similar) and five PCs onto the logical workspace.

2. IP Plan: Define a Class C private IP network: 192.168.1.0 with a Subnet Mask of 255.255.255.0 (or /24).
3. Assign Host IDs: Assign unique host addresses to each PC:
 - PC0: 192.168.1.10
 - PC1: 192.168.1.11
 - PC2: 192.168.1.12, and so on.

B. Physical Wiring (Cabling)

1. Select the Copper Straight-Through cable tool.
2. Connect each PC's FastEthernet0 port directly to the next available Fa0/x port on the Switch.
3. Wait for the link lights on the connections to turn green, confirming the physical link is active.

C. Logical Configuration (IP Addressing)

1. Click on PC0 → Desktop tab → IP Configuration.
2. Select Static.
3. Enter the planned IP Address (192.168.1.10) and Subnet Mask (255.255.255.0).
4. Repeat this process for all remaining PCs (PC1, PC2, etc.), ensuring each has its unique, incremented IP address.

D. Verification and Command Execution

1. Click on PC0 → Desktop tab → Command Prompt.
2. **Verify Local Settings (ipconfig):**
 - Execute: ipconfig
 - *Note the output (IP Address and Subnet Mask) to confirm the static assignment.*
3. **Test Connectivity (ping):**
 - Execute: ping 192.168.1.11 (Pinging PC1 from PC0).
 - *Note the output. Successful connectivity is confirmed by four Reply from... messages and 0% loss.*
4. **Hardware Address (Getmac):**
 - Execute: Getmac
 - *Note the physical MAC Address of the NIC on PC0.*
5. **Address Resolution Cache (arp):**

- Execute: `arp -a`
- *Note the output. The ARP cache should now show the IP and MAC address of PC1, which was learned during the successful ping.*

6. Local Machine Name (hostname):

- Execute: `hostname`
- *Note the output (should be PC0).*

CN-PRACTICAL 2: IPV4 SUBNETTING PROOF

Aim: To demonstrate and verify the logical segmentation of a physical network into two distinct subnets using the Subnet Mask 255.255.255.192 and to observe the resulting communication boundaries in a Star Topology.

- i. The objective of this practical is to visually verify that the specific Subnet Mask of 255.255.255.192 successfully segments a single physical network into two logical subnets, thereby blocking communication between them.
- ii. **Begin the physical setup** by placing one **Switch** (e.g., a 2960 model) and four **PCs** (labeled PC0, PC1, PC2, PC3) onto the Packet Tracer workspace.
- iii. Next, establish the **Star Topology** by using the **Copper Straight-Through** cable to connect the **FastEthernet0** port of every PC to the next available port on the Switch (starting with Fa0/1).
- iv. Confirm the connections are active by observing that all link status indicators turn green.
- v. Once the physical connections are complete, proceed to the **Logical Configuration** by assigning IP addresses according to the subnetting plan.
- vi. **First, configure PC0**, navigating to the **Desktop** tab and then **IP Configuration**, setting the IP Address to 192.168.1.10 and the Subnet Mask to 255.255.255.192. **Next, configure PC3** with the IP Address 192.168.1.20 and the same Subnet Mask (255.255.255.192).
- vii. These two PCs now belong to **Subnet 1** (192.168.1.0).
- viii. **Then, configure PC1** with the IP Address 192.168.1.70 and the Subnet Mask 255.255.255.192.
- ix. **Finally, configure PC2** with the IP Address 192.168.1.80 and the Subnet Mask 255.255.255.192. PC1 and PC2 are now logically segmented into **Subnet 2** (192.168.1.64).
- x. The final stage is **Verification and Results** using the Command Prompt on **PC0**.
- xi. **First, verify PC0's settings** by executing the command ipconfig; the output should show the IP Address 192.168.1.10 and the mask 255.255.255.192.
- xii. **Second, test communication within Subnet 1** by executing ping 192.168.1.20 (pinging PC3); the **Expected Result** is **Success (0% Loss)**, proving the mask recognizes them as local.
- xiii. **Third and most importantly, test communication between subnets** by executing ping 192.168.1.70 (pinging PC1 in Subnet 2); the **Expected Result** is **Failure (100% Loss)**.
- xiv. This failure demonstrates that PC0's subnet mask tells the PC that PC1 is on a different network, causing the packet to be dropped because no Router is present.

- xv. This practical confirms that the **Subnet Mask explicitly defines the network boundaries** and controls communication, overriding the physical connection.

CN-PRACTICAL 3: CONFIGURATION AND VERIFICATION OF STATIC IP ADDRESS

Aim: To configure and verify a Static IPv4 address on a host device within a Local Area Network (LAN) using a network simulation tool.

Step 1: To execute this practical, start by placing a single PC and a Switch on the workspace and connecting them using a Copper Straight-Through cable.

Step 2: Open the PC configuration by clicking on the device and navigating to the Desktop tab, then clicking on IP Configuration.

Step 3: You will see two options: DHCP and Static. Select the Static radio button; this tells the computer to ignore any automatic assignments and only use the numbers you provide. In the IPv4 Address field, enter 192.168.1.50, and click the Subnet Mask field to let it auto-fill as 255.255.255.0.

Step 4: To verify that the address is truly "Static," close the configuration windows and open the Command Prompt on the same PC.

Step 5: Type the command ipconfig and press Enter to see your assigned address.

Step 6: To prove the address is permanent, you can delete the cable, wait a few seconds, and reconnect it; when you run ipconfig again, the address 192.168.1.50 will still be there. This proves that a Static IP is a permanent manual assignment that does not change unless an administrator manually edits it.

CN-PRACTICAL 4: CONFIGURATION AND VERIFICATION

RIPV2 DYNAMIC ROUTING

Aim: The primary objective of this lab is to **design, configure, and verify a multi-node network** that utilizes **RIPv2 (Routing Information Protocol version 2)** to enable communication between two disparate Local Area Networks (LANs).

Phase 1: Building the Physical Topology

Open Cisco Packet Tracer and drag the following components onto the workspace:

- 1. Routers:** Grab two 2911 Routers. (Name them R1 and R2).
- 2. Switches:** Grab two 2960 Switches.
- 3. End Devices:** Grab two PCs. (Name them PC-A and PC-B).
- 4. Connections:** Connect PC-A to Switch0 (Copper Straight-Through).
 - Connect Switch0 to R1 GigabitEthernet 0/0 (Copper Straight-Through).
 - Connect R1 GigabitEthernet 0/1 to R2 GigabitEthernet 0/1 (Copper Crossover).
 - Connect R2 GigabitEthernet 0/0 to Switch1 (Copper Straight-Through).
 - Connect Switch1 to PC-B (Copper Straight-Through).

Phase 2: Assigning IP Addresses to PCs

Before the routers can talk, the computers need their "home" addresses.

1. Click PC-A > Desktop > IP Configuration:

- IP Address: 192.168.10.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.10.1 (This is the router's interface).

2. Click PC-B > Desktop > IP Configuration:

- IP Address: 192.168.20.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.20.1

Phase 3: Configuring Router Interfaces

We must turn on the router ports and give them IPs. We will use the CLI (Command Line Interface).

Interface tab.

On Router R1:

Router> enable

```
Router# configure terminal
# Set IP for the Local Network (LAN)
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.10.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
# Set IP for the Connection to the other Router (WAN)
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.10.10.1 255.255.255.252
Router(config-if)# no shutdown
```

On Router R2:

```
Router> enable
Router# configure terminal
# Set IP for the Local Network (LAN)
Router(config)# interface gigabitEthernet 0/0
Router(config-if)# ip address 192.168.20.1 255.255.255.0
Router(config-if)# no shutdown
Router(config-if)# exit
# Set IP for the Connection to the other Router (WAN)
Router(config)# interface gigabitEthernet 0/1
Router(config-if)# ip address 10.10.10.2 255.255.255.252
Router(config-if)# no shutdown
```

Phase 4: Configuring RIP

Right now, PC-A cannot ping PC-B because R1 doesn't know where the 192.168.20.0 network is. RIP will teach it.

On Router R1:

```
Router(config)# router rip
Router(config-router)# version 2
```



```
Router(config-router)# network 192.168.10.0
Router(config-router)# network 10.10.10.0
Router(config-router)# no auto-summary
```

On Router R2:

```
Router(config)# router rip
Router(config-router)# version 2
Router(config-router)# network 192.168.20.0
Router(config-router)# network 10.10.10.0
Router(config-router)# no auto-summary
```

Phase 5: Verification (Testing)

1. Wait 30 seconds (RIP sends updates every 30 seconds).
2. On R1, type: show ip route.
 - Students should see a line starting with R 192.168.20.0. This "R" stands for RIP.
3. Go to PC-A > Command Prompt and type: ping 192.168.20.2.
4. If the ping is successful, the routing is working!

CN-PRACTICAL 5: OSPF CONFIGURATION AND VERIFICATION

Aim: To configure and verify OSPF (Open Shortest Path First) dynamic routing between two routers in a multi-network environment using Cisco Packet Tracer

1. Network Topology Setup

- **Devices used:** 2 Routers (1841 series) and 2 PCs.
- **Connections:** * PC0 to Router0
 - Router0 to Router1
 - Router1 to PC1

2. IP Addressing Scheme

Assign the following IP addresses:

- **Network 1 (PC0 side):** 192.168.1.0/24 (PC0: .2, Router0 FastEthernet 0/0: .1)
- **Network 2 (Between Routers):** 192.168.2.0/24 (Router0 FastEthernet 0/1: .1, Router1 FastEthernet 0/0: .2)
- **Network 3 (PC1 side):** 192.168.3.0/24 (Router1 FastEthernet 0/1: .1, PC1: .2)

3. Basic Configuration

- **PC Configuration:** Set IP addresses and Default Gateways for PC0 and PC1 via the Desktop > IP Configuration tab.
- **Router Interface Configuration:** Enter the Router's **Config** tab to assign IP addresses to specific interfaces and turn the status to **ON**
- **Verification:** A quick ping test is performed between the PC and its connected Router to ensure the link is active

4. OSPF Routing Configuration

This is the core of the practical. Use the CLI (Command Line Interface) for OSPF.

Concepts Introduced:

- **Area:** In this small setup, both routers are placed in Area 0.
- **Wildcard Mask:** The instructor explains that a Wildcard Mask is the "opposite" of a Subnet Mask (e.g., for 255.255.255.0, the wildcard is 0.0.0.255).

Router 0 Configuration Commands:

Router> enable

Router# configure terminal

Router(config)# router ospf 1

Router(config-router)# network 192.168.1.0 0.0.0.255 area 0

Router(config-router)# network 192.168.2.0 0.0.0.255 area 0

Router 1 Configuration Commands:

Router> enable

Router# configure terminal

Router(config)# router ospf 2

Router(config-router)# network 192.168.3.0 0.0.0.255 area 0

Router(config-router)# network 192.168.2.0 0.0.0.255 area 0

Testing and Verification

- The instructor uses the **Simple PDU (Message tool)** in Packet Tracer to test end-to-end connectivity.
- The status shows **Success** when sending a packet from PC0 to PC1, confirming that OSPF routing is working correctly.

CN-PRACTICAL 6: BGP CONFIGURATION AND VERIFICATION

Aim: To design, configure, and verify **External Border Gateway Protocol (eBGP)** connectivity across three distinct **Autonomous Systems (AS 1000, 2000, and 3000)** to enable end-to-end communication between isolated Local Area Networks (LANs).

1. Network Topology & IP Planning

The setup consists of **3 Autonomous Systems** connected in a line:

- **AS 1000 (Router 0):** Connected to LAN 192.168.1.0/24.
- **AS 2000 (Router 1):** Connected to LAN 192.168.2.0/24.
- **AS 3000 (Router 2):** Connected to LAN 192.168.3.0/24.
- **Serial Links:**
 - Between AS 1000 and AS 2000: 10.0.0.0/8
 - Between AS 2000 and AS 3000: 20.0.0.0/8

2. Initial Configuration

- Assign IP addresses to all PC desktops (IP, Subnet Mask, and Default Gateway).
- Configure the Router interfaces (GigabitEthernet for LANs and Serial for WANs) and ensure all interfaces are not shut down.
- **Verification:** At this stage, pinging between different networks will fail because the routers do not yet have routes to the distant networks.

3. BGP Configuration on Routers

To enable BGP, you must define the local AS number, advertise local networks, and manually define neighboring routers.

Router 0 (AS 1000):

```
Router> enable
```

```
Router# configure terminal
```

```
Router(config)# router bgp 1000
```

```
Router(config-router)# network 192.168.1.0
```

```
Router(config-router)# network 10.0.0.0
```

```
Router(config-router)# neighbor 10.0.0.2 remote-as 2000
```

Router 1 (AS 2000):

This router acts as a bridge and has two neighbors.

```
Router(config)# router bgp 2000
```

```
Router(config-router)# network 192.168.2.0
```

```
Router(config-router)# network 10.0.0.0
```

```
Router(config-router)# network 20.0.0.0
```

```
Router(config-router)# neighbor 10.0.0.1 remote-as 1000
```

```
Router(config-router)# neighbor 20.0.0.2 remote-as 3000
```

Note: As soon as the first neighbor is configured, the CLI will log that the BGP adjacency has moved to "Up"

Router 2 (AS 3000):

```
Router(config)# router bgp 3000
```

```
Router(config-router)# network 192.168.3.0
```

```
Router(config-router)# network 20.0.0.0
```

```
Router(config-router)# neighbor 20.0.0.1 remote-as 2000
```

4. Verification of Connectivity

- Go to the **Command Prompt** of a PC in AS 1000 (e.g., 192.168.1.2).
- Ping a PC in the furthest network, AS 3000 (e.g., 192.168.3.2).

CN-PRACTICAL 7: CONFIGURE IPV6 ON CISCO ROUTER

Aim: To design a network topology in Cisco Packet Tracer and perform basic IPv6 configuration, including enabling unicast routing, assigning link-local and global unicast addresses to router interfaces, and verifying connectivity between hosts using SLAAC (Stateless Address Autoconfiguration).

Introduction:

IPv6 is the sixth version of the IP protocol. It consists of eight groups of four hexadecimal digits. IPv6 is a 128-bit address having an address space of 2^{128} .

Steps to Configure IPv6 on Cisco Packet Tracer:

Step 1: First, open the Cisco Packet Tracer desktop and create a network topology, something like the image, and an IPv6 addressing table given below.

Step 2: Configuring the GigabitEthernet Interfaces.

- First, we will configure the **GigabitEthernet0/0** using CLI.
- Click on **router0** and go to CLI and type the commands given below:

```
Router>enable
```

```
Router#configure terminal
```

Enter configuration commands, one per line. End with CNTL/Z.

```
Router(config)#ipv6 unicast-routing
```

```
Router(config)#int Gig0/0
```

```
Router(config-if)#ipv6 address FE80::1 link
```

```
Router(config-if)#ipv6 address FE80::1 link-local
```

```
Router(config-if)#no shutdown
```

- Now we will configure the **GigabitEthernet0/1** Interface.
- Click on **router0** and go to CLI and type the commands given below:

```
Router(config-if)#int Gig0/1
```

```
Router(config-if)#ipv6 address FE80::1 link-local
```

```
Router(config-if)#no shutdown
```

- Then, both the interfaces will be active now:

Step 3: Configuring IPv6 address in both Interfaces using CLI:

- CLI commands to configure IPv6 address in GigabitEthernet0/0 and GigabitEthernet0/1 ports are given below:

Router# enable

Router# configure terminal.

Enter configuration commands, one per line. End with CNTL/Z.

Router(config)#int Gig0/0

Router(config-if)#ipv6 address 2001:DB8:AAAA:A::1/64

Router(config-if)#no shutdown

Router(config-if)#int Gig0/1

Router(config-if)#ipv6 address 2001:DB8:AAAA:B::1/64

Router(config-if)#no shutdown

Router(config-if)#

Step 4: We have configured the router. Now, change the settings of hosts in IPv6 configuration:

- First, click on PC0 and go to the desktop, then IP configuration.
- Now find the IPv6 configuration.
- Change the settings from static to automatic, and then after a few seconds, the IPv6 address and default gateway are displayed.
- Similarly, repeat this procedure with PC1 and PC2

Step 5: Now we have to verify the connection by pinging the IPv6 address of PC0 in PC1.

- First, click on PC1 and go to the command prompt, and type ping < IPv6 address>
- command: ping 2001:DB8:AAAA:A:230:A3FF:FE0E:8B9C

CN-PRACTICAL 8: USING WIRESHARK NETWORK ANALYZER TO CAPTURE AND ANALYZE ICMP, TCP, HTTP, UDP, AND FTP PROTOCOLS

Aim: To capture and analyze various network protocol packets (ICMP, TCP, HTTP, UDP, and FTP) using the Wireshark Network Analyzer tool.

Requirements

- PC / Laptop
- Internet connection
- **Wireshark installed**
- Command Prompt / Terminal
- Web Browser (Chrome / Firefox)

Step 1: Start Wireshark

1. Open **Wireshark**
2. Select the active network interface
 - Ethernet (for LAN)
 - Wi-Fi (for wireless)
3. Click **Start Capturing Packets**

A) ICMP Protocol (Ping)

Step 2: Generate ICMP Traffic

1. Open **Command Prompt**
2. Type: **ping google.com**
1. Press Enter

Step 3: Apply ICMP Filter

- In the Wireshark filter bar, type: **Icmp**
- Press Enter

Observation:

- ICMP Echo Request

- ICMP Echo Reply

B) TCP Protocol

Step 4: Generate TCP Traffic

1. Open any website (example: google.com)
2. Keep capture running

Step 5: Apply TCP Filter

- tcp

Observation:

- TCP 3-way handshake
 - SYN
 - SYN-ACK
 - ACK

C) HTTP Protocol

- Important: HTTP works only on **non-HTTPS** sites.

Step 6: Generate HTTP Traffic

1. Open browser
2. Visit:
3. http://example.com

Step 7: Apply HTTP Filter

http

Observation:

- GET request
- HTTP 200 OK response

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D) UDP Protocol

Step 8: Generate UDP Traffic

1. Open Command Prompt
2. Run:
3. nslookup google.com

Step 9: Apply UDP Filter

- udp

Observation:

- DNS request and response over UDP

E) FTP Protocol

- FTP uses **TCP port 21**

Step 10: Generate FTP Traffic

1. Open Command Prompt
2. Type: ftp ftp.dlptest.com
3. Login with:
 - **Username:** dlpuser
 - **Password:** rNrKYTX9g7z3RgJRmxWuGHbeu

Step 11: Apply FTP Filter

- ftp

Observation:

- USER command
- PASS command
- FTP responses